

Synthometricon: A search engine to identify conceptual overlap in measures in the behavioral sciences

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Online supplement:

- Preprint: <https://osf.io/preprints/psyarxiv/nfgmv>
- GitHub: <https://github.com/synth-science/synthometricon/>
- App: <https://huggingface.co/spaces/magnolia-psychometrics/synthometricon>

Abstract:

Every month, scores of new measures and constructs enter the behavioral science literature. Researchers struggle to keep an overview of their subfields, let alone the wider field, with an estimated number of allegedly different constructs now exceeding 30,000. Proposing constructs is easy, finding redundancies and overlaps is not. The effort of finding redundancies using participant responses grows quadratically with the number of measures. This cost motivates the use of language models, which can simplify and speed up the process without requiring new data collection. Earlier work employed latent semantic analysis (LSA) to make use of the textual nature of questionnaires in the social and behavioral sciences. However, LSA treats sentences as bags of words and ignores word order, syntax, and negation, all of which shape how respondents interpret items. Recent advances demonstrate that transformer-based sentence embedding models offer a more viable solution for predicting empirical relatedness between survey items. In the presented work, we use a fine-tuned language model, the SurveyBot3000, to introduce the

Synthometricon, an open-access web application for systematic detection of conceptual overlap. The application indexes over 570,000 items across more than 97,000 scales from approximately 28,000 instruments in the APA PsycTests database and other sources.

Introduction

Behavioral scientists propose new constructs and measures faster than the field can establish how they relate to one another. When Cronbach and Meehl (1955) envisioned the nomological network to map out the relationships between psychological constructs and their corresponding manifestations in the physical world, they noted that scientists have the freedom to expand the network, if an observation will not fit into it. Today, researchers expand the network readily but rarely show first that an observation did not fit. More than 30,000 supposedly novel constructs have been proposed, and new ones appear every month (Anvari et al., 2025; Elson et al., 2023). This proliferation abandons the conservatism Cronbach and Meehl advocated, fragments the field and hinders the accumulation of knowledge. Among the most problematic consequences of a fragmented research field are jingle-jangle fallacies: cases where researchers unwittingly study the same construct under different names (jangle fallacies; e.g., grit and conscientiousness, Credé et al., 2017) or conflate distinct constructs under a single label (jingle fallacies; e.g., mindfulness, Altgassen et al., 2024). Disciplinary silos exacerbate the fragmentation, as subfields propose constructs and measures in isolation from one another.

Conventional approaches to evaluating construct redundancy cannot keep pace with this proliferation. For example, the Synthetic Aperture Personality Assessment (SAPA; Condon & Revelle, 2015) administers random subsets of items from various personality scales to survey respondents. With growing sample size, SAPA-data can effectively be used to evaluate the (non-)redundancy of measures and proposed constructs. Unfortunately, this approach doesn't scale well, as including additional measures requires a quadratic increase in survey participants. To obtain a pairwise N of 100 for correlations between 570,000 items in a SAPA study in which each participant answers 100 items, we would need more than 3.2 billion participants.

Given the textual nature of surveys, questionnaires, checklists, tests and other forms of assessment in the social and behavioral sciences, natural language processing holds the potential to address the inconsistent mapping between measures and constructs. Early works in the field have demonstrated that survey item semantics are related to survey responses (e.g., Arnulf et al., 2014; Larsen & Bong, 2016) using latent semantic analysis (LSA; Deerwester et al., 1990). In essence, LSA maps text data (i.e., survey items) into a semantic space by identifying latent topics that underlie word co-occurrences across documents (i.e., scales). Numerical vector representations of items and scales can then be obtained and compared using distance metrics, such as cosine similarity. Building on LSA,

Rosenbusch et al. (2020) developed the Semantic Scale Network, an online tool to automatically detect semantic overlap between psychological scales. The application allows researchers to input items from a new scale and receive quantifications of semantic similarity with over 4,000 existing scales in its corpus, without requiring empirical data collection. The Semantic Scale Network first showed the potential of using natural language processing to detect potentially redundant measures in the scientific literature. However, a central limitation arises from the reliance on word co-occurrences of LSA-based approaches. As noted by Rosenbusch and colleagues, semantic similarity between two conceptually overlapping scales may be low if the items of each scale are phrased differently (e.g., “I’m disapproving and judgmental about my own flaws and inadequacies” and “I worry a lot”). Similarity estimates between scales may likewise appear inflated, if two items measuring different constructs employ similar wording (e.g., homonyms, as in “I am the life of the party” and “I vote for the same party in every election”) or if words combine into expressions whose meaning differs from that of their parts (e.g., “I often butt-dial friends” [clumsiness] and “I often booty-call friends” [sociosexuality]). In that sense, LSA-based approaches may fail to challenge jingle-jangle fallacies due to surface level characteristics of the item text.

Moving beyond the constraints of LSA, transformer models have emerged as the current state-of-the-art in many natural language processing applications (Vaswani et al., 2017; see Hussain et al., 2023 for an accessible introduction). In contrast to LSA, transformer-based embedding models produce contextualized n-dimensional vector representations of text, with each token (i.e., word or sub-word) encoded according to its role within a particular sentence. For example, the aforementioned usage of the word “party” would yield different embeddings depending on the context of each item. Rather than embedding individual tokens, sentence transformer models can be used to represent entire text sequences as a unified vector (Reimers & Gurevych, 2019). Current research demonstrates the predictive power of these embeddings for understanding survey responses. At the item level, Abdurahman et al. (2023) extrapolated participant ratings to novel personality items using sentence embeddings. At the aggregate level, researchers have trained models to produce item embeddings to predict pairwise correlations between items and scales, as well as reliability coefficients with high accuracy (Hernandez & Nie, 2023; Hommel & Arslan, 2025; Schoenegger et al., 2025; Wulff & Mata, 2025).

In summary, sentence embeddings offer a solid basis for detecting overlap among measures, because they a) capture contextualized meaning in text beyond shared wording, b) remain robust to paraphrasing and syntactic variations, c) can be produced to reflect the

empirical association between survey scales, and d) scale efficiently to thousands of instruments without requiring additional data collection.

In this work, we introduce Synthometricon: a web application powered by the SurveyBot3000, a language model developed by Hommel & Arslan (2025). Synthometricon enables systematic detection of conceptual overlap across more than 28,000 psychological measures from the APA PsycTests database and other sources.

The Synthometricon search engine

We release Synthometricon as an open-access web application on Hugging Face Spaces: <https://search.synth-science.org/>¹.

The web application functions as a conceptual search engine, currently allowing researchers and practitioners to explore 28,233 instruments extracted from the APA PsycTests database, with permission of the American Psychological Association, and other sources (Rosenbusch et al., 2020; Larsen et al., 2025). In contrast to conventional search engines which adopt keyword-based approaches, Synthometricon relies on sentence embeddings of items, scale labels and instrument names to match user queries with database records. The application operates in two different modes: item-based search and label-based search.

In item-based search, users enter one or more items into the search interface. Items can come from existing scales, scales under development, or be formulated ad hoc to explore a construct of interest. We recommend this mode for detecting conceptual overlap, as it operates on the SurveyBot3000 model, which was explicitly trained to predict pairwise correlations between items and scales (Hommel & Arslan, 2025).

The utility of item-based search is exemplified by the results depicted in Figure 1. The example shows item queries plausibly related to narcissism, with reverse-coded items appropriately flagged (Figure 1a). The resulting matches (Figure 1b) reveal substantial conceptual overlap with multiple instruments measuring dark personality traits. Notably, Synthometricon identified high similarity with the Short Dark Tetrad (Paulhus et al., 2020), particularly its subscale labeled "special". Here, Paulhus and colleagues used a euphemistic label in place of the term narcissism. This illustrates how focusing on item content rather

¹ Updated links will be maintained at <https://github.com/synth-science/synthometricon> in case of migration. The web application's source code is freely available at <https://huggingface.co/spaces/magnolia-psychometrics/synthometricon/tree/main>.

than scale labels allows Synthometricon to identify substantive overlap that is missed by conventional search methods.

In label-based search, users enter a construct, scale or instrument name (see Figure 1c). This mode uses a smaller, general purpose model and produces results which reflect semantic overlap, which is less likely to correspond to empirical relatedness in measures.

Figure 1. The Synthometricon web application interface.

a)

Search by Items Labels

Type an item, press Enter...

- 1 Receiving compliments is important to me.
- 2 I rarely doubt my decisions or actions.
- 3 I'm just like everybody else. reversed
- 4 I'm skilled at persuading others to believe what I say.
- 5 Being humble doesn't suit my personality.
- 6 I consider myself to be truly exceptional.
- 7 Gaining recognition and status is important to me.
- 8 I'm destined for remarkable achievements.

Try an example Clear

Press Enter to add an item, paste a list to add several at once, and end a line with **(R)** to mark reverse scoring.

b)

Search by Items Labels

Type an item, press Enter...

Press Enter to add an item, paste a list to add several at once, and end a line with **(R)** to mark reverse scoring.

28,307 INSTRUMENTS Copy link Export CSV

Filter instrument names. Absolute similarity (high to low) 5 / page Filters

.669 Short Dark Tetrad 28 items • 4 scales • 2021

APA PsycTests 2021 4 scales English 1 warning

Short Dark Tetrad (28 items)

- Crafty (7 items) .481
- Special (7 items) .295
- Wild (7 items) .669
- Mean (7 items) .088
- Mean (7 items) .184

Extracted scale count differs from the APA PsycTests record
Item text unverified — no source PDF
[Report a problem with this record](#)

c)

Search by Items Labels

Grandiosity

Name a scale or trait, or describe the construct in a few words, then press Enter.

28,307 INSTRUMENTS Copy link Export CSV

Filter instrument names. Absolute similarity (high to low) 5 / page Filters

.746 Daily Personality Pathology Manifestations Measure 30 items • 34 scales • 2016

.743 Narcissistic Vulnerability Scale 48 items • 5 scales • 2005

APA PsycTests 2005 5 scales English 1 warning

Narcissistic Vulnerability Scale (48 items)

- Grandiosity (4 items) .574
- Exploitation (11 items) .743
- Self-Esteem Regulation (18 items) .228
- Idealization (2 items) .161
- Parents (13 items) .236
- Parents (13 items) .252

Extracted item text may differ from the source PDF
Scale count unverified — no catalogue record
[Report a problem with this record](#)

Note. The example shows how to use Synthometricon to search for instruments related to narcissism in a) item-based search, and c) label-based search.

Below, we detail the development of Synthometricon, including its underlying models, algorithms and database.

Sentence embedding models

To transform the extracted items and scale labels into searchable representations, we employed two distinct sentence embedding models: The SurveyBot3000 for item-based search and the all-MiniLM-L6-v2 for label-based search.

The SurveyBot3000 builds on the all-mpnet-base-v2 sentence transformer model, which Hommel and Arslan (2025) fine-tuned through a two-stage training protocol to predict pairwise item correlations. In the first stage, the model was calibrated to distinguish semantically opposing concepts, ensuring items measuring opposite construct poles (e.g., extraversion versus introversion) produce negatively correlated embeddings. This calibration was necessary because standard sentence transformers typically yield positive cosine similarities even for contradictory statements, as their high-dimensional representations encode multiple correlated linguistic features. In the second stage, the model was fine-tuned on 74,339 item pairs from various questionnaires using a cosine similarity loss objective. Here, the model learned to predict item relatedness from the text in the item stems by minimizing the discrepancy between predicted and empirical correlations. Initial validation on a hold-out sample demonstrated strong convergence between synthetic and empirical correlations for item pairs, scale pairs, and Cronbach's alpha coefficients. A subsequent pre-registered validation study confirmed generalization to new content domains, achieving out-of-sample accuracies of $r = .59$ [95% CI 0.58;0.60] for item correlations, $.84$ [0.82;0.87] for scale correlations, and $.84$ [0.79;0.90] for reliability coefficients. The average prediction error for scale correlations was $RMSE = 0.16$ [0.14;0.17]. For our use case, it is most important that high positive and high negative correlations between scales have low prediction error, so that similar and opposite measures can be found. For synthetic predictions with high predicted absolute scale correlations, RMSE was lower (Hommel & Arslan, 2025). Conversely, scales that are strongly correlated empirically are often predicted to be less strongly related, so users who want to find all related measures need to screen beyond the top results (see Recommendations). Given these predictive capabilities, we use the SurveyBot3000 to estimate correlations between scales in the APA PsycTests data and thereby identify potential conceptual overlap.

In addition to the SurveyBot3000 model, Synthometricon uses the smaller all-MiniLM-L6-v2² model to power label-based search. Here, in contrast to item-based search, scale-representations in Synthometricon are constructed by computing 384-dimensional embeddings from scale labels and instrument names.

Employing both models allows researchers to probe Synthometricon for different kinds of potential overlap between search queries and existing measures in the database on both the conceptual and on the semantic level.

Data sources

Synthometricon indexes instruments from various data sources, with the PsycTests database currently serving as the primary source. Maintained by the American Psychological Association, PsycTests constitutes one of the largest collections of measures in the social and behavioral sciences, currently containing over 83,000 records³ with approximately 300 new records added monthly. Each record includes metadata such as constructs measured, instrument type, associated references, and publication year. Where available, records also include the actual test materials and supporting documentation. In addition to PsycTests, Synthometricon integrates complementary instruments obtained from the Semantic Scale Network (Rosenbusch et al., 2020), the ALIGNS-Project (Larsen et al., 2025) and a targeted search for online materials (see below).

PsycTests corpus

We obtained a PsycTests database snapshot on August 21, 2023, which contained 71,692 records, 32,720 of them with PDF documents of test materials. Documents in PsycTests consist of unstructured content, combining textual and image data from scanned pages. Processing the documents therefore required text recognition (e.g., converting bitmaps in poor resolutions to text) as well as natural language understanding capabilities (e.g., identifying individual items and scale structures in the document). Both challenges can be addressed using modern vision-language models (VLMs). While traditional optical character recognition (OCR) techniques may incorporate contextual information, they do so typically in shallow, local forms such as lexicons, or character n-gram models. By contrast, transformer-based vision-language models interpret visual input within a larger context. Contextual processing enables the model to compensate for obscured characters or poor image quality

² <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2> (retrieved April 1, 2025)

³ 83,479 records according to <https://www.apa.org/pubs/databases/psyc-tests> (retrieved September 16th, 2026)

by making use of surrounding document-level semantic information and broad prior knowledge and thereby improving extraction accuracy for degraded or low-quality documents. To ensure the copyright integrity of the test materials and adhere to open-science best practices (Schroeders et al., 2026), we used Gemma-4-31B (4-bit quantized; Abd et al., 2026), which represented the state-of-the-art in open-weights models at the time of writing. We employed the llama.cpp framework (Gerganov, 2026) to efficiently handle model inference locally. We distributed the total computational workload of approximately 960 GPU-hours across ten NVIDIA Tesla V100 instances, each with 32 GB VRAM, on the scientific computing infrastructure of Leipzig University. Because these GPUs have less memory, production used a context window of 32,768 tokens; the consistency and validity evaluations (see below) ran on a local GPU with a window of 165,536 tokens, using the ggml-org conversion of the same quantized model.

Figure 2 illustrates the simplified VLM-assisted information extraction pipeline. For each document, pages containing instrument material were rendered to images and jointly passed to Gemma-4-31B in combination with three different prompt instructions to extract item-, scale-, and meta-data (see Supplementary Note 1 for further details). We used grammar-constrained decoding (Geng et al., 2023) to reliably coerce model output into a structured data format. We then passed the rendered document and the preceding model output back to the VLM to map extracted items to their corresponding scales and determine item polarity. This process produced a complete instrument tree that can accommodate the majority of nested measure hierarchies typically found in psychological instruments. Extraction failed for 52 documents (e.g., because of timeouts on excessively long documents), and 56 documents contained no items that matched our extraction instructions, leaving 32,612 documents with items. Overall, this initially yielded 711,837 items (Mdn = 15, IQR = 8–27 items per document; Mdn = 5, IQR = 3–11 items per scale) and 113,287 scales (Mdn = 3, IQR = 1–5 per document).

Additional data sources

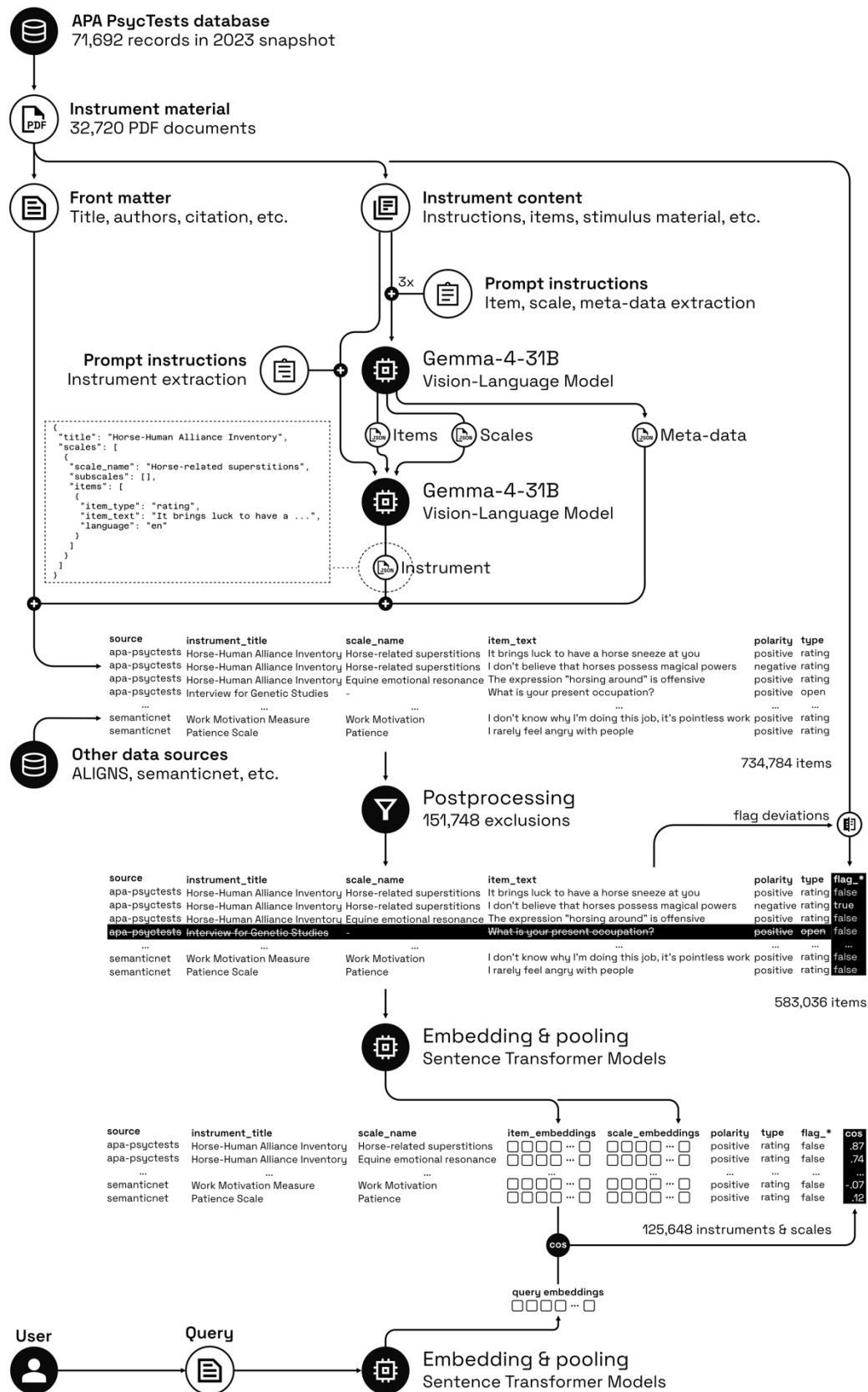
We obtained three additional datasets to complement the final Synthometricon corpus. The Semantic Scale Network (Rosenbusch et al., 2020) provided a set of 111 single-scale instruments totaling 2,836 items (Mdn = 20, IQR = 12–31 items per document). We furthermore integrated 21 instruments containing 605 items (Mdn = 14, IQR = 12–20 items per document; Mdn = 10, IQR = 6–14 items per scale) and 41 scales (Mdn = 1, IQR = 1–2 per document) from Larsen et al. (2025). Finally, to close remaining gaps in the coverage of Synthometricon, we ran an orchestrated multi-agent search using Claude Sonnet 5 and

Haiku 4.5 to extract items from the primary literature and the web for widely used instruments (per PsycInfo usage count). It yielded 120 instruments with 3,176 items (Mdn = 18.5, IQR = 10–30.5 items per document; Mdn = 5, IQR = 1–11 items per scale) and 557 scales (Mdn = 4, IQR = 1–6 per document). The additional data source incorporation is documented in more detail in Supplementary Note 6. The Semantic Scale Network and ALIGNS do not record reverse keying, and web-retrieved sources did so only in part; items without keying information are pooled as positively keyed (see Limitations).

Postprocessing

The combined Synthometricon corpus comprises 32,916 instruments holding 734,784 items (718,454 of them unique). We then removed the 52 failed extractions and, applying the following filters sequentially and counting items, excluded 105,142 items that used non-rating scale formats, as the SurveyBot3000 model was specifically trained on rating scales. We further pruned 12,251 items lacking text content (e.g., semantic differentials with only labeled endpoints), 2,584 items exceeding 250 characters, and 3,593 items incorporating visual stimulus material. Finally, meta-data inferred during the extraction process (see Supplementary Note 1) led us to remove 1,813 non-scorable items (e.g., demographic variables), 1,192 items from clinical intake forms, and 25,173 items measuring abilities (e.g., cognitive ability tests). This left a final corpus of 28,233 instruments totaling 583,036 items (570,561 unique; Mdn = 15, IQR = 8–25 items per document; Mdn = 5, IQR = 3–11 items per scale) and 97,415 scales (Mdn = 3, IQR = 1–5 per document). Postprocessing removed 4,683 instruments entirely, among them 19 of the 120 web-retrieved instruments, 17 because their items use other response formats (e.g., the Beck Depression Inventory) and 2 because their items exceed 250 characters; these instruments are not searchable.

Figure 2. Overview of the Synthometricron data extraction and embedding pipeline



Note. Overview of the multi-stage Synthometricon data pipeline. Source documents are parsed with a VLM (Gemma-4-31B) to extract structured instrument hierarchies and metadata. Sentence transformer models generate pooled vector representations at the instrument and scale level. User queries are embedded via the same models, with cosine similarity between query and target vectors determining conceptual relatedness.

Embedding and pooling

We constructed a hierarchical embedding system that represents instruments at the global level as well as on each subsequent level of nesting. In doing so, we did not take into account whether the instrument measured a coherent construct (e.g., extraversion) or comprised theoretically independent constructs (e.g., a Big Five instrument). The specific embedding calculation procedure varied depending on the search mode.

In item-based search, we began by encoding each individual item with the SurveyBot3000, producing 768-dimensional vector representations. The item-level embeddings were then aggregated to produce instrument- and scale-level representations through mean pooling.

Conventional scale analysis requires reverse-coding negatively keyed items before computing scale scores. Likewise, SurveyBot3000 embeddings of opposite-pole items correlate negatively and must be reversed before aggregation. To this end, we first unit-normalize all individual item embeddings. We then directly negate the vectors of items explicitly marked as reverse-coded in the original instrument. Because $\cos(-v, w) = -\cos(v, w)$, the mathematical negation precisely mirrors traditional reverse-scoring in the embedding space, i.e., changing the sign of the item's predicted correlations while maintaining their magnitude. We finally compute the unweighted mean of these aligned vectors to produce a single, aggregated centroid representation of the scale. The python implementation of this algorithm can be found in the online supplemental section (see Supplementary Note 2).

Label-based search drew on a simpler approach. We used the all-MiniLM-L6-v2 model to separately encode the instrument name (obtained from PsycTests metadata) and the scale name (transcribed by Gemma-4-31B), then averaged these embeddings (for scales with subscales, including the subscale names) to produce a single 384-dimensional vector representation.

Scoring algorithm

Identifying conceptually related documents through embedding-based retrieval, commonly termed semantic search (Reimers & Gurevych, 2019), constitutes a standard application of sentence transformers. The retrieval mechanism embeds documents (i.e., instruments and scales) and search queries into a shared semantic space. For the SurveyBot3000 model, this space approximates relationships between constructs. Relatedness is then quantified using cosine similarity, which captures the angular proximity between embeddings. Cosine similarity is simply a correlation coefficient computed across the dimensions of the two embeddings rather than across respondents (it equals the Pearson correlation exactly when each embedding is mean-centered).

In item-based search, users specify their query by entering survey items that purportedly measure a construct of interest, along with keying information specifying each item's polarity. Following the embedding procedure described previously, these query items are encoded using SurveyBot3000 and aggregated through polarity-aligned mean pooling (see Supplementary Note 2). Synthometricon then computes the cosine similarity between the resulting query embedding and all instrument- and subscale-level embeddings in the database. Scales exhibiting the highest absolute cosine similarity are therefore likely to measure constructs conceptually related to the queried scale.

Label-based search follows the same retrieval logic. Here, users provide a single keyword describing a construct, scale, or instrument name, which is embedded via all-MiniLM-L6-v2 and compared to the corresponding scale embeddings in the database.

Data quality

The overall utility of Synthometricon depends on both the predictive accuracy of the underlying embedding model and the quality of the corpus. As the predictive performance of the SurveyBot3000 model already has been determined in prior research by Hommel & Arslan (2025), our evaluation here focuses primarily on the faithfulness of the VLM-assisted extraction process. Errors in the data may stem from external factors inherent to the original database (e.g., clerical errors in database records) or processing errors introduced during automated extraction. We necessarily limit our evaluation to the latter, as database records must constitute the ground truth for assessing extraction accuracy. Specifically, we evaluated both the consistency (i.e., whether multiple extraction attempts produce the same structured output) and the validity (i.e., the congruence between VLM-extracted instruments

and human-coded standards) of the extraction process. For both purposes we randomly selected 100 documents from the APA PsycTests database snapshot.

In our consistency evaluation, we ran each of the four extraction steps for five repeated extractions, resulting in ten pairwise comparisons. In addition to the inherently non-deterministic nature of (visual) language models (e.g., Schroeders et al., 2026), we used the recommended settings for probabilistic sampling with Gemma-4-31B⁴ and therefore expected the output to vary. Re-sampling models and evaluating consistency is oftentimes used in natural language processing to determine robustness and validity (Farquhar et al., 2024). We therefore considered the item-extraction step to be successful if a) the total item count across each iteration agreed for each document, and b) Krippendorff's α for the extracted item texts was $\geq .90$, using the squared normalized Levenshtein distance between texts as the difference function (Krippendorff, 2004; Levenshtein, 1966). Similarly, our criteria for successful scale-extractions were a) count agreement on scales and subscales, as well as b) the tree similarity of scale hierarchies between repeated extractions ($\geq .90$). In meta-data extractions, we assessed consistency by unanimous agreement of VLM-inferred judgements about the instrument (e.g., is the measure a medical intake form). Finally, we assessed the consistency of item-to-scale mappings at the instrument level. Because this extraction step depends on the output of previous item and scale extractions, we held those inputs fixed. Each instrument prompt received the same human-validated set of items and scale hierarchy for a document. We then measured the consistency of VLM mappings a) by Krippendorff's α for agreement of item-to-scale assignment ($\geq .90$), and b) the share of item-scale pairings with unanimous polarity flags (i.e., whether items were marked as reverse-coded; $\geq .90$). See Supplementary Note 3 for details on these evaluation metrics. Fifteen of the 100 documents (15%, 95% CI [9%, 23%]) failed at least one of these criteria (item extraction: 3, scale extraction: 11, item-to-scale mapping: 3, meta-data: 0).

In validity evaluation, we compared the full instrument trees resulting from VLM-extraction for the 100 documents to the expected structures as defined by trained human coders (see Supplementary Note 4 for details). Because manually transcribing a complete document is exceptionally labor-intensive (single instrument tree oftentimes comprises hundreds of lines of text), human coders recorded only the most important attributes of each instrument (e.g., item texts, scale hierarchy, reverse-keyings). In total, 73 extractions (73%, 95% CI [64%,

⁴ <https://huggingface.co/unsloth/gemma-4-31B-it-qat-GGUF>; evaluation runs: <https://huggingface.co/ggml-org/gemma-4-31B-it-GGUF> (retrieved May 23rd, 2026)

81%]) matched the human-coded fixtures (texts at a normalized Levenshtein similarity of at least .80, ignoring case). In 11 further documents, the discrepancies were due to errors in the human coding, and two documents returned truncated output in this run: one was extracted correctly in all five consistency runs, and the other differed from the human coding only because the coders had merged multi-part questions into single items. We thoroughly examined the remaining 14 extractions (14%, 95% CI [9%, 22%]) and qualitatively classified their discrepancies as reported in Supplementary Note 5. We conclude that the majority of these cases arise from ambivalent, incomplete, or erroneous information in the respective documents. While language models are known to occasionally confabulate plausible sounding but factually incorrect responses, (commonly termed "hallucinations"; e.g., Huang et al., 2025; Liu et al., 2024), we found no evidence of hallucinated items in our evaluation. Keying disagreed with the human coding for a single item (Supplementary Note 5). In some cases, Gemma-4-31B misspelled named entities (e.g., "Ross- Miro[w]sky Neighborhood Disorder Scale") or slightly paraphrased items without altering their meaning (e.g., "[...] my voice varies *throughout* the day" versus "[...] my voice varies *during* the day."). To test how much the remaining discrepancies affect search results, we embedded the human-coded and machine-extracted versions of the audited documents and compared the resulting scale vectors (Supplementary Note 8). Of 223 human-coded scales, 222 (99.6%, 95% CI [97.5%, 99.9%]) were represented in the audited machine extraction, and across all pairs of these scales, cosines computed from machine-extracted text correlated $r = .986$, 95% CI [.975, .996], with those computed from human-coded text (RMSE = .025, 95% CI [.014, .036]). The search index holds a separate extraction run; for the audited documents it contains, the correlation was $r = .968$, 95% CI [.927, .994] (RMSE = .044, 95% CI [.019, .078]), mostly because two documents lost their subscale structure in that run.

In addition to evaluating the validity of machine-coded instrument trees, we implemented additional metrics that flag potential inconsistencies between extracted data and source materials. These heuristic checks serve as sensitive signals for users to exercise additional scrutiny, rather than as definitive indicators of poor data quality. Synthometricon reports a) divergence between the number of items extracted by gemma-4-31B and item counts documented in PsycTests metadata, b) discrepancies in scale counts between extraction and metadata specifications, c) substantial deviations in extracted item text from the raw PDF content, identified through fuzzy string matching, and d) if items were machine-translated by gemma-4-31B during extraction.

We deliberately implemented conservative heuristics that prioritize sensitivity over specificity. By accepting a higher rate of false positives, we minimize the risk of overlooking genuine extraction errors and ensure that Synthometricon users are alerted even when the extraction may in fact be correct but metadata incomplete or ambiguous. A cross-check against our validity evaluation bears this out: of the 73 audited documents that are part of the searchable corpus, 34 carried at least one flag, but only 9 of these (26%, 95% CI [15%, 43%]) had a genuine discrepancy in the indexed extraction, and the flags caught 9 of the 15 genuine discrepancies among these documents (60%, 95% CI [36%, 80%]). Item count discrepancies affected 5,795 instruments, 21.0% of the 27,650 instruments whose PsycTests record documents an item count. Deviations in item count can arise from various reasons such as incomplete or inaccurate metadata documentation, model hallucinations, or failure to parse complex document layouts. Scale counts were documented for only 4,122 instruments; of these, 1,508 (36.6%) deviated from the extracted scale count. We assessed verbatim fidelity by matching each extracted item with the substring in the source PDF that best aligned it. Between these two strings we then calculated the Levenshtein distance and normalized it by the length of the longer string. Instruments whose PDFs had no usable text layer were excluded: fewer than 200 characters of text ($n = 713$), or items printed as images (at least one image per page beyond the PsycTests logo and fewer than 500 characters of text per page; $n = 2,365$). Across the remaining 492,030 items, 65.1% matched verbatim and 11.6% fell below 50% ($Mdn = 100\%$, $IQR = 96.3\text{--}100\%$; $M = 88.6\%$, $SD = 23.0\%$). Overall, 8,361 of 24,907 checkable instruments (33.6%) contained at least one item below the 95% threshold. Such cases can arise when source documents contain partly scanned pages or malformed layouts (e.g., table elements disrupting sequential item order). Low similarity can also result from model rephrasing or hallucinations. Translations of one or more items were included in 6.0% of the instruments. Consequently, 13,086 instruments (46.4%) were flagged for additional user scrutiny. We think the search results will be enhanced by these flags, providing the user with additional information on data quality to facilitate informed decisions.

Discussion

We introduce the Synthometricon to provide researchers and practitioners with an automated approach to locating measures within an increasingly sprawling ecosystem that has outgrown manual cataloging efforts. In contrast to conventional methods that rely on empirical data, Synthometricon evaluates the relatedness of measures based on the textual information contained in items, scale and instrument names. The use of natural language

processing to identify overlapping measures has precedent in tools like the Semantic Scale Network (Rosenbusch et al., 2020). Synthometricon contributes several key improvements to this line of work. First, it employs sentence transformer models rather than LSA, producing contextualized embeddings that reduce spurious matches driven by coincidental wording and superficial linguistic aspects (Reimers & Gurevych, 2019). Second, its use of the SurveyBot3000 model extends the capabilities of baseline sentence transformers in identifying conceptual relatedness, as the model is explicitly trained to approximate empirical relationships between measures (Hommel & Arslan, 2025). Third, by incorporating a large portion of the APA PsycTests database, Synthometricon covers the most comprehensive and authoritative collection of psychological measures that we are aware of.

Recommendations

Synthometricon has two modes of search. The scale-label-based search essentially allows researchers to find potentially useful scales quickly. However, labels can mislead about content, because of jingle-jangle fallacies, and in our case also because some of the scale labels had to be inferred by our model. Essentially, this functionality extends the keyword-based search for scales that the APA PsycTests database and similar products already support.

By contrast, the item-content-based search is the novel possibility and the main way in which we envision the Synthometricon search adding value. We see two main use cases.

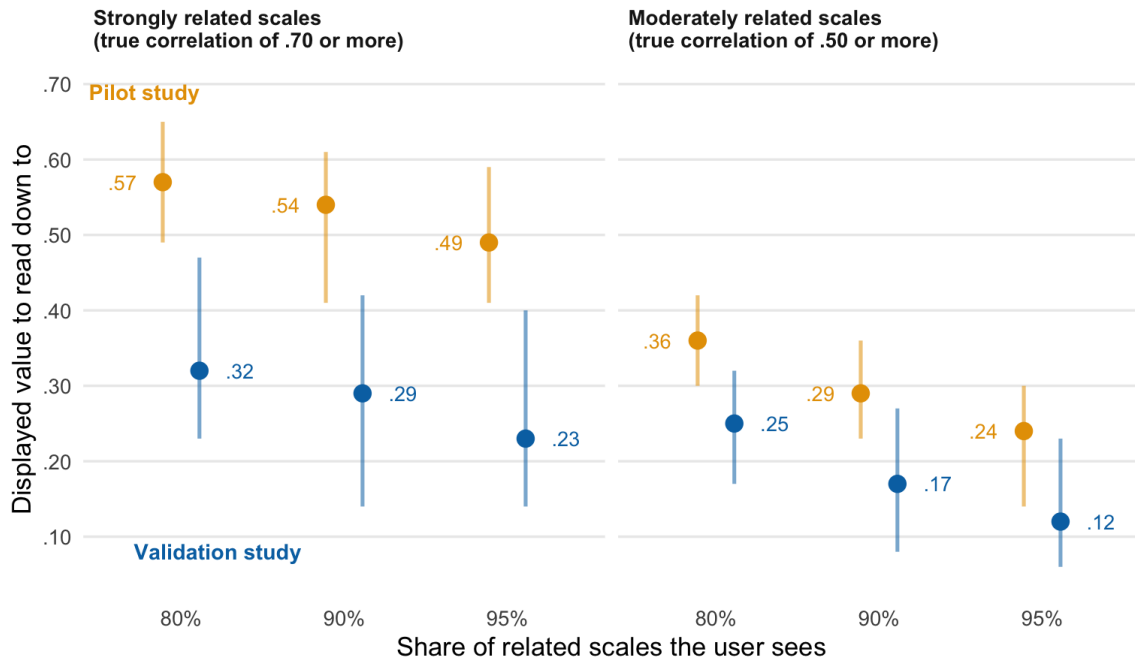
First and most straightforwardly, instead of developing a new measure, researchers can produce items tapping the construct of interest and find existing work quickly, even if the construct has been labelled differently by previous researchers. We recommend searching with at least three items, ideally five or more. In our reanalysis of the SurveyBot3000 validation data, single items reached an accuracy of .72, 95% CI [.64, .78], and five items .84, 95% CI [.77, .90], compared with .83 to .86 for the full scales they were drawn from (Supplementary Note 7), and accuracy for scale pairs with at least five items each was .88 (Hommel & Arslan, 2025). Reverse-keyed items should be marked: when half of them were left unmarked, accuracy fell from .83, 95% CI [.75, .89], to .74, 95% CI [.64, .82] (Supplementary Note 7). The search engine surfaces relevant results and researchers use their expertise, their human judgment and potentially empirical data collection to ascertain whether any of the found measures are fit for purpose.

Second, sometimes researchers may want to find out whether a new measure is indeed novel. They are less interested in finding a single good match but rather want to demonstrate a lack of matches. Evidence of absence is of course a harder problem. For one, although we believe our corpus is the largest currently searchable, it is still incomplete. In addition, the synthetic scale correlations do not perfectly capture empirical associations. If researchers wish to find 80% of the scales in the corpus that are truly correlated at least .70 with their own scale, they cannot screen only down to synthetic correlations of .70. They have to go lower to account for prediction error in the synthetic estimates to about .57, 95% CI [.49, .65], for scales like those in the SurveyBot3000 pilot study, but to about .32, 95% CI [.23, .47], for scales like those in the validation study (Figure 4; Supplementary Note 7). We show two numbers because the SurveyBot3000 pilot study, which sampled a more homogeneous set of scales from personality psychology and related subdisciplines, implies different cutoffs than the validation study, which sampled more heterogeneous scales (Hommel & Arslan, 2025). Even though the screening burden may still be immense for many queries, we think this is still a large improvement to screening all measures, which is clearly infeasible, or using label-based search, which will systematically miss the same content under different labels. Researchers must nevertheless guard against false reassurance as the absence of a matching instrument in Synthometricon does not preclude the existence of a relevant measure elsewhere in the literature.

Figure 4. How far down the result list users need to read

How far down the result list to read to see most related scales

A lower value means reading further down the list. For scales like those in the validation study, users had to read much further down than for well-known personality questionnaires.



Data: SurveyBot3000 pilot study (22 personality and well-being questionnaires, N = 493) and validation study (57 scales sampled from APA PsycTests, N = 387). Points: median over 1,000 draws of the true correlations; lines: 95% bootstrap confidence intervals. About 45 strongly and 180 to 430 moderately related scale pairs per study.

Note. Displayed value (synthetic correlation) down to which users need to read to see 80%, 90% or 95% of the scales that truly correlate at least .70 (left) or .50 (right) with their query, in the SurveyBot3000 pilot study (22 personality and well-being questionnaires, N = 493) and validation study (57 scales sampled from APA PsycTests, N = 387). Medians over 1,000 draws of the true correlations (Supplementary Note 7).

When used in these two ways, Synthometricon supports taxonomic refinement in two ways. Prospectively, it can prevent redundant work from being done. Retrospectively, it enables systematic flagging of scale pairs for potential redundancy in the existing literature. Importantly, Synthometricon cannot and should not replace empirical investigations of construct validity. It provides a highly useful heuristic for locating conceptually overlapping instruments within an otherwise intractable literature, but synthetic matches do not constitute indirect evidence of a measure's validity. Cases where Synthometricon indicates high overlap with preexisting scales would require submitting researchers to justify the novelty of their measure through theoretical arguments and, ideally, empirical evidence of discriminant validity. We agree with Rosenbusch et al. that tools like Synthometricon serve best as decision support systems, providing evidence to inform rather than replace expert judgment (see also Lin, 2025).

Artificial intelligence, particularly language modeling techniques, is increasingly transforming (meta-)scientific practice. Concerns have been raised about potential threats to scientific integrity (Binz et al., 2025) and the risk of amplifying existing problems, such as a focus on Western convenience samples, through unreflective use (Cummins, 2025). However, responsible human-in-the-loop applications can enhance research quality by addressing systemic problems, such as the tragedy of the commons engendered by the asymmetry in costs between adding measures to the pile and sorting through it. We think AI tools work best when they are used to reduce information overload. As another example, Cummins et al. (2026) proposed using language models to automatically flag deviations between preregistrations and submitted manuscripts, drawing reviewers' attention to potential discrepancies without replacing human judgment. Since human reviewers almost never check preregistrations against submissions, the AI, instead of replacing human judgment, is used to scaffold it, by highlighting potential mismatches for expert review. We envision a similar role for Synthometricon in the peer review process. Currently, human authors, reviewers and editors do not perform an exhaustive content search of existing psychometric instruments when evaluating allegedly novel additions. They go on their memory and perhaps keyword search. By surfacing a shorter list of potentially related measures, Synthometricon makes the human expert's task easier. Again, the tool does not replace judgment but scaffolds it and can help experts design better studies of construct validity.

Limitations

Synthometricon's results depend fundamentally on data quality. Despite considerable effort to ensure reliable automated extraction from APA PsycTests materials, errors remain inevitable. However, our comparisons of machine- and human-coded instruments indicate that such errors are rare and barely change the scale representations the search relies on (Supplementary Note 8). While we implemented sensitive quality metrics to alert users to potential issues, they cannot substitute for careful scrutiny of individual search results. Because source materials were poorly structured and often omitted information about nested scale structures, our data may occasionally miss capturing scales at some level of nesting. This poses problems for search queries targeting hierarchically organized instruments, such as some Big Five inventories. For example, if users enter item queries that cover a broader measure of conscientiousness, they will be matched against a global scale (i.e., “personality”) and subscales (e.g., “orderliness”) for instruments that reported subfacets. As

a consequence, estimated overlap may appear deflated when the conceptual breadth of a query misaligns with the granularity of retrieved scales.

In addition, not all of the measures in APA PsycTests have their item texts documented; only 39.4% of the 71,692 records in our 2023 snapshot are searchable (28,251 records, including those whose items we supplemented from additional sources; see Supplementary Note 6), although these records account for 54.4% of documented uses. And not all measures that exist are documented in APA PsycTests, although coverage is broad. As one of our goals is to reduce fragmentation in psychology (Anvari et al., 2025), we will need to fill in the white areas of the map (see Figure 3). As mentioned in the previous section, the absence of substantial overlap between the user query and the Synthometricon search results does not guarantee that a similar measure does not already exist in the behavioral science literature.

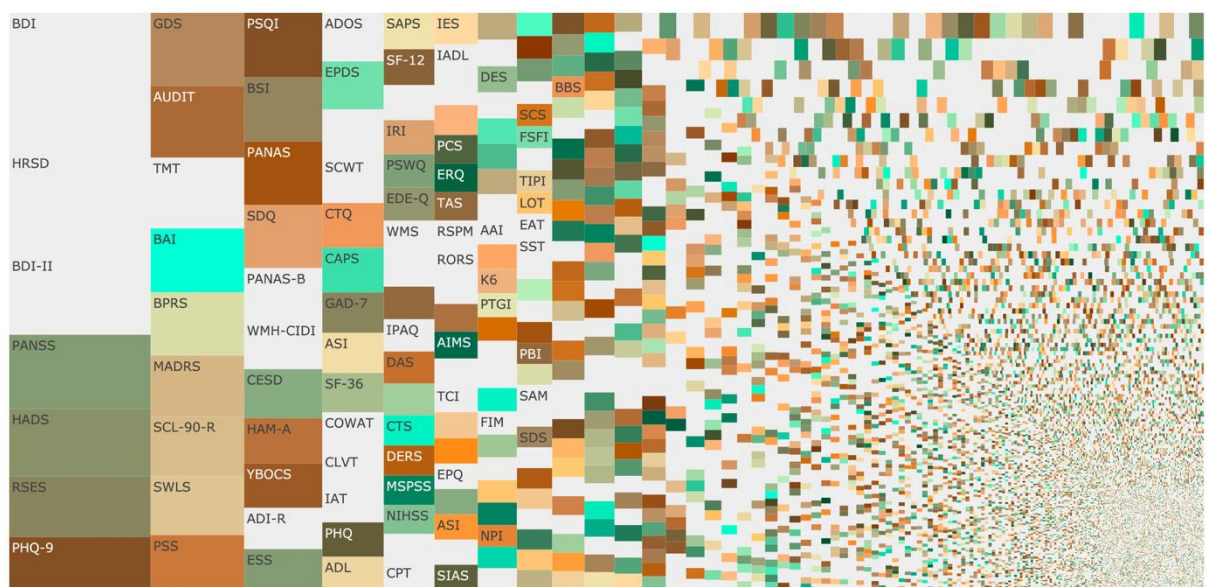


Figure 3. A treemap plot showing psychological measures in APA PsycTests. Each tile represents a measure. Tiles are sized proportionally to how often measures were documented to be used by APA coders in the PsycInfo database (publications up to 2023). Only the measures shown in color can be found with item-based search, i.e., at least one of their rating-scale items is in the search index; grey measures lack item text or were excluded during postprocessing (e.g., ability tests, interviews, and yes/no or multiple-choice formats). This treemap does not show tests that had no usage recorded in PsycInfo ($n = 40,574$).

Beyond data quality, Synthometricon's performance is naturally bounded by the SurveyBot3000 model's predictive capabilities, which, while strong, cannot achieve perfect

accuracy. As reported in Hommel & Arslan (2025), the model's prediction errors occur asymmetrically, with negative item correlations occasionally misclassified as positive. In general, when predicting correlations between scales, the model's prediction error (RMSE) compares to the sampling error observed in empirical studies with sample sizes of about $N = 20$ for positive and $N = 80$ for negative synthetic correlations.

Some additional minor constraints warrant mention. First, given the SurveyBot3000's limited multilingual capabilities, non-English instruments were machine-translated and may yield less accurate search results. Second, because Synthometricon lacks continuous integration with PsycTests, newly published instruments are not included in its database. Third, items from the Semantic Scale Network and ALIGNS, and some web-retrieved items, carry no keying information and are pooled as if positively keyed; leaving reverse-keyed items unmarked lowers accuracy considerably (Supplementary Note 7). Finally, legal constraints limit the information we can provide to users. We cannot display individual item text and must present results only at the aggregate scale or instrument level. These same constraints prevent us from linking to original publications, restricting us to providing DOI references to the corresponding APA PsycTests records.

Outlook and future research

The Synthometricon data provides a strong foundation that can be expanded both temporally and in content breadth. Our August 2023 data snapshot can be updated to incorporate the steady stream of new measures published since then, ensuring researchers can screen for novelty against the most current literature. Extraction took about 960 GPU-hours for 32,720 documents, i.e., less than two GPU-minutes per document, so the roughly 300 records added to PsycTests each month could be processed in about 9 GPU-hours. Beyond temporal expansion, coverage can extend to sources not fully represented in APA PsycTests, such as instruments found in supplementary materials, preregistration platforms, open science repositories, and adjacent behavioral science disciplines. Since its inaugural release, we have added 5,550 items from 233 instruments from the Semantic Scale Network, the ALIGNS item corpus and a targeted web search to the search database. Integration with large-scale data aggregation initiatives, such as the Item Response Warehouse (Domingue et al., 2025) and data donation platforms (e.g., Reichmann & Hommel, n. d.), would enable continuous database expansion and enhanced coverage.

Conclusion

A fundamental tension in the behavioral sciences is balancing researchers' freedom to propose new constructs and measures against the need to prevent unchecked proliferation that fragments the field. Currently, it is easier to add a measure to the pile than sort through it. In a classic tragedy of the commons, researchers gain individually from adding to the pile, the community bears the cost of the resulting fragmentation, and nobody has much incentive to tidy up. Reducing this asymmetry in costs by making it easier to sort through the pile, to find and link existing work, is therefore key. Previous tools were limited in their usefulness by data coverage and sorting ability. Synthometricon improves on both counts, offering researchers a practical means to navigate the existing measurement landscape before adding to it. When researchers can efficiently map their work onto what already exists, fewer will opt to add further measures to the pile and readers can draw connections more freely.

Recent proposals (Elson et al., 2023; Lin, 2026) go further and propose audits during review to prevent the publication of redundant or mislabeled measures and envision an open measure repository with machine-readable metadata against which language models assess overlap. Synthometricon shows that this screening layer is feasible now. Because it needs only item text, it works on the literature as it stands; no structured reporting standards are needed first. Assembling our database showed why such standards would still help. Where discrepancies in data extraction occurred, they were driven mostly not by the technical limitations of the Gemma-4-31B model, but by missing source data and the erratic formatting of unstructured PDFs. For now, Synthometricon is decision support for authors, reviewers, and editors. As databases grow and become better structured, we anticipate a shift from isolated measure development toward collaborative curation of an increasingly integrated nomological network, a *living* standard that is reassessed as new measures and evidence appear (Anvari et al., 2025; Lin, 2026).

Acknowledgments

We thank Hannes Rosenbusch for sharing his data from the Semantic Scale Network to complement our database. We also thank Marie Rosenstock and Miguel Bilen for contributing human-coded instruments for data quality evaluation. We thank the American Psychological Association (APA) for granting permission to use the APA PsycTests Database for this research.

Data availability

The study relies on test materials obtained from the APA PsycTests database. In accordance with the licensing agreement with the American Psychological Association, we are not permitted to share the raw PDF documents or make item-level content publicly available. We provide the processing code and all reproducible components that can be shared under the terms of our agreement. Researchers with appropriate access to APA PsycTests can reproduce the dataset by applying our extraction and preprocessing scripts to their own licensed copy.

Code availability

We have shared the code for data extraction and preprocessing at <https://github.com/synth-science/synthometricon>. The source code for Synthometricon is shared at <https://huggingface.co/spaces/magnolia-psychometrics/synthometricon/tree/main>.

Author contributions

B.E.H.: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Data Curation, Writing - Original Draft, Visualization, Supervision, Project administration, Funding acquisition.

A.I.K.: Validation, Formal analysis, Data Curation, Writing - Review & Editing, Visualization

R.C.A.: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

All authors approved the final manuscript for submission.

Competing interests

Björn E. Hommel is affiliated with magnolia psychometrics GmbH, a private consulting agency which has agreed to maintain the Synthometricon web application (currently hosted on Hugging Face) on a voluntary, non-remunerated basis, without incurring legal liability.

There are no competing interests, financial or otherwise, related to this research. Potential

future commercial applications may be developed. As a long-term governance plan, the authors have decided to release all non-copyrighted assets on the ZPID PsychArchive in case magnolia psychometrics loses interest in maintaining the app.

Funding

This work was funded by the German Research Foundation (Deutsche Forschungsgemeinschaft, DFG; grant number 546323839).

Prior versions

A previous version of this manuscript was posted as a preprint at <https://osf.io/preprints/psyarxiv/nfgmv>. The present article is a substantially revised version of the same work.

Ethics

This research analyzed test documentation and previously published data from the SurveyBot3000 studies and collected no data from human participants, so it required no ethics approval.

Open practices

None of the analyses reported here were preregistered. We report how we determined the size of our audit sample, all data exclusions and all measures; the study involved no experimental manipulations.

Use of generative AI

Besides the uses described in the Methods and in Supplementary Note 6, the authors used Claude Code (Anthropic) to write and review code, to check the reported numbers against the analysis outputs and find errors, and to suggest wording for parts of the manuscript and the response to reviewers. The authors reviewed all outputs and take full responsibility for the content of this article.

Supplemental Material

Supplementary Notes 1–8 are provided as Supplemental Material.

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